

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418900
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Zhang; Guodong et al.

Methods for transmission time interval bundling in the uplink

Abstract

Methods, systems, and apparatus for supporting uplink Transmission Time Interval (TTI) bundling in Long Term Evolution (LTE) are provided. Methods, systems, and apparatus for signaling, activation/deactivation, and wireless transmit/receive unit (WTRU) behavior are also provided. In an example, a downlink control information (DCI) associated with a repeated physical uplink shared channel (PUSCH) transmission may be received. A hybrid acknowledgment repeat request (HARQ) process information may be determined using the received DCI. A repeated PUSCH transmission may be sent in accordance with one or more time units, the repetition configuration, and the HARQ process information.

Inventors: Zhang; Guodong (Woodbury, NY), Wang; Jin (Princeton, NJ), Pan; Kyle Jung-Lin (Saint James, NY), Gauvreau; Jean-Louis (La Prairie, CA), Terry; Stephen E. (Northport, NY)

Applicant: INTERDIGITAL PATENT HOLDINGS, INC. (Wilmington, DE)

Family ID: 1000008818242

Assignee: INTERDIGITAL PATENT HOLDINGS, INC. (Wilmington, DE)

Appl. No.: 18/392461

Filed: December 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240172195 A1	May. 23, 2024

Related U.S. Application Data

continuation parent-doc US 17750617 20220523 US 11895632 child-doc US 18392461
continuation parent-doc US 16854825 20200421 US 11343816 20220524 child-doc US 17750617
continuation parent-doc US 16156054 20181010 US 10674505 20200602 child-doc US 16854825

continuation parent-doc US 14851257 20150911 US 10104655 20181016 child-doc US 16156054
continuation parent-doc US 12421838 20090410 US 9172509 20151027 child-doc US 14851257
us-provisional-application US 61083789 20080725
us-provisional-application US 61081444 20080717
us-provisional-application US 61052725 20080513
us-provisional-application US 61044111 20080411

Publication Classification

Int. Cl.: **H04W72/12** (20230101); **H04L1/00** (20060101); **H04L1/1607** (20230101); **H04L1/1812**
(20230101); **H04L1/1867** (20230101); **H04L5/00** (20060101); **H04W72/0446**
(20230101); **H04W72/20** (20230101)

U.S. Cl.:

CPC **H04W72/0446** (20130101); **H04L1/0003** (20130101); **H04L1/1621** (20130101);
H04L1/1635 (20130101); **H04L1/1812** (20130101); **H04L1/189** (20130101);
H04L5/0048 (20130101); **H04W72/12** (20130101); **H04W72/20** (20230101);

Field of Classification Search

CPC: H04L (1/0025); H04L (1/189); H04L (1/0038); H04L (1/0035); H04L (1/0039); H04L
(1/0029); H04L (5/0035); H04L (5/0055); H04L (5/0057); H04L (12/2869); H04L
(12/5695); H04L (47/00); H04L (47/70); H04L (47/78); H04L (47/821); H04L (47/827);
H04W (72/0406); H04W (72/04)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7996029	12/2010	Chun et al.	N/A	N/A
9172509	12/2014	Zhang et al.	N/A	N/A
2007/0275086	12/2006	Nowak et al.	N/A	N/A
2008/0186892	12/2007	Damnjanovic	N/A	N/A
2008/0232284	12/2007	Dalsgaard	370/310	H04W 72/23
2009/0086657	12/2008	Alpert et al.	N/A	N/A
2009/0129317	12/2008	Che et al.	N/A	N/A
2009/0196366	12/2008	Shen et al.	N/A	N/A
2009/0197630	12/2008	Ahn et al.	N/A	N/A
2009/0204862	12/2008	Chun et al.	N/A	N/A
2009/0238091	12/2008	Kim et al.	N/A	N/A
2009/0238121	12/2008	Kotecha	N/A	N/A
2009/0238128	12/2008	Park et al.	N/A	N/A
2009/0245225	12/2008	Tseng	N/A	N/A
2010/0023830	12/2009	Wengertter	370/479	H04L 1/0025
2010/0080152	12/2009	Lindh	370/312	H04W 72/0446
2010/0173659	12/2009	Shin	455/500	H04L 1/0073
2010/0192035	12/2009	Sagfors	714/E11.131	H04L 1/1812
2010/0195641	12/2009	Tsai	370/352	H04M 7/0075

2010/0202382	12/2009	Park et al.	N/A	N/A
2010/0275086	12/2009	Bergquist et al.	N/A	N/A
2010/0279628	12/2009	Love	455/70	H04L 5/0091
2010/0279634	12/2009	Sagfors	455/127.1	H04L 1/0038
2010/0290509	12/2009	Dalsgaard et al.	N/A	N/A
2010/0322122	12/2009	Synnergren et al.	N/A	N/A
2010/0325508	12/2009	Hu et al.	N/A	N/A
2011/0002293	12/2010	Yuk et al.	N/A	N/A
2011/0021228	12/2010	Kim et al.	N/A	N/A
2011/0083066	12/2010	Chung et al.	N/A	N/A
2011/0124360	12/2010	Sagfors	455/509	H04W 28/18
2011/0141878	12/2010	Che	370/216	H04L 1/0025
2011/0173519	12/2010	Kuri et al.	N/A	N/A
2011/0182245	12/2010	Malkamaki et al.	N/A	N/A

OTHER PUBLICATIONS

3rd Generation Partnership Project (3GPP), R1-080339, "On VoIP Uplink Coverage for LTE", Ericsson, TSG-RAN WG1 #51bis, Sevilla, Spain, Jan. 14-18, 2008, 3 pages. cited by applicant

3rd Generation Partnership Project (3GPP), R1-080444, "On the Need for VoIP Coverage Enhancement for the E-UTRA UL", Alcatel-Lucent, 3GPP TSG-RAN WG1 #51-bis, Sevilla, Spain, Jan. 14-18, 2008, pp. 1-10. cited by applicant

3rd Generation Partnership Project (3GPP), R1-081102, "Outcome of Offline Discussions on PDCCH Contents", Ericsson, 1 page. cited by applicant

3rd Generation Partnership Project (3GPP), R1-081103, "Reply LS on Uplink Coverage for LTE", RAN1, TSG-RAN WG1 # 52, Sorrento, Italy, Feb. 11-15, 2008, 3 pages. cited by applicant

3rd Generation Partnership Project (3GPP), R2-081185, "UL Coverage Enhancement for VoIP Transmission", Philips, NXP Semiconductors, 3GPP TSG RAN WG2 Meeting #61, Sorrento, Italy, Feb. 11-15, 2008, 3 pages. cited by applicant

3rd Generation Partnership Project (3GPP), R2-081326, "Reply LS on Uplink Coverage for LTE", RAN1, 3GPP TSG RAN WG2 Meeting #61, Sorrento, Italy, Feb. 11-15, 2008, 3 Pages. cited by applicant

3rd Generation Partnership Project (3GPP), R2-081446, "RAN2 Aspects of the Solutions for Subframe Bundling", Alcatel-Lucent, 3GPP TSG-RAN WG2 #61bis, Shenzhen, China, Mar. 31-Apr. 4, 2008, 8 pages. cited by applicant

3rd Generation Partnership Project (3GPP), R2-081465, "Evaluation of TTI-Bundling Alternatives", Ericsson, 3GPP TSG-RAN WG2 #61bis, Shenzhen, China, Mar. 31-Apr. 4, 2008, pp. 1-4. cited by applicant

3rd Generation Partnership Project (3GPP), R2-081466, "Text Proposal for TTI Bundling", Ericsson, 3GPP TSG-RAN WG2 #61bis, Shenzhen, China, Mar. 31-Apr. 4, 2008, pp. 1-8. cited by applicant

3rd Generation Partnership Project (3GPP), R2-081779, "EUL Coverage Enhancements", Ericsson, 3GPP TSG-RAN WG2 #61bis, Shenzhen, China, Mar. 31-Apr. 4, 2008, pp. 1-3. cited by applicant

3rd Generation Partnership Project (3GPP), R2-081799, "Report of MAC Activities", MAC Rapporteurs (Qualcomm Europe, Ericsson), 3GPP TSG-RAN WG2 #61bis, Shenzhen, China, Mar. 31-Apr. 4, 2008, pp. 1-2. cited by applicant

3rd Generation Partnership Project (3GPP), R2-082148, "TTI-Bundling Considerations for TDD", Ericsson, 3GPP TSG-RAN WG2 #62, Kansas City, USA, May 5-9, 2008, pp. 1-7. cited by applicant

3rd Generation Partnership Project (3GPP), R2-082149, "Text Proposal for TTI Bundling", Ericsson, 3GPP TSG-RAN WG2 #62, Kansas City, USA, May 5-9, 2008, pp. 1-6. cited by

applicant

3rd Generation Partnership Project (3GPP), R2-082253, "TTI Bundling Impact on DRX Inactivity Timer", ZTE, 3GPP TSG-RAN WG2#62, Kansas City, USA, May 5-9, 2008, pp. 1-2. cited by applicant

3rd Generation Partnership Project (3GPP), R2-082317, "TTI Bundling Configuration", CATT, 3GPP TSG RAN WG2 #62, Kansas City, USA, May 5-9, 2008, pp. 1-7. cited by applicant

3rd Generation Partnership Project (3GPP), TS 36.213 V8.2.0, "Technical Specification Group Radio Access Network, Evolved Universal Terrestrial Radio Access (E-UTRA), Physical Layer Procedures (Release 8)", Mar. 2008, pp. 1-30. cited by applicant

3rd Generation Partnership Project (3GPP), TS 36.300 V8.2.0, "Technical Specification Group Radio Access Network, Evolved Universal Terrestrial Radio Access and Evolved Universal Terrestrial Radio Access Network, (E-UTRAN), Overall Description, Stage 2 (Release 8)", Sep. 2007, pp. 1-109. cited by applicant

3rd Generation Partnership Project (3GPP), TS 36.321 V8.0.0, "Technical Specification Group Radio Access Network, Evolved Universal Terrestrial Radio Access (E-UTRA) Medium Access Control (MAC) Protocol Specification (Release 8)", Dec. 2007, pp. 1-23. cited by applicant

Primary Examiner: Liu; Jung

Attorney, Agent or Firm: Condo Roccia Koptiw LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/750,617, filed May 23, 2022, which is a continuation of U.S. patent application Ser. No. 16/854,825, filed Apr. 21, 2020, which issued as U.S. patent Ser. No. 11/343,816, which is a continuation of U.S. patent application Ser. No. 16/156,054, filed Oct. 10, 2018, which issued as U.S. Pat. No. 10,674,505 on Jun. 2, 2020, which is a continuation of U.S. patent application Ser. No. 14/851,257, filed Sep. 11, 2015, which issued as U.S. Pat. No. 10,104,655 on Oct. 16, 2018, which is a continuation of U.S. patent application Ser. No. 12/421,838, filed Apr. 10, 2009, which issued as U.S. Pat. No. 9,172,509 on Oct. 27, 2015, which claims the benefit of U.S. Provisional Application No. 61/044,111, filed Apr. 11, 2008; 61/052,725, filed May 13, 2008; 61/081,444, filed Jul. 17, 2008; and 61/083,789, filed Jul. 25, 2008 the entire contents of all of which are incorporated herein by reference.

FIELD OF INVENTION

(1) This application is related to wireless communications.

BACKGROUND

(2) Third generation partnership project (3GPP) and 3GPP2 are considering long term evolution (LTE) for radio interface and network architecture. There is an ever-increasing demand on wireless operators to provide better quality voice and high-speed data services. As a result, wireless communication systems that enable higher data rates and higher capacities are a pressing need.

(3) In evolved Universal Mobile Telecommunication System (UMTS) Terrestrial Radio Access (evolved UTRA), the network architecture includes a radio access network that provides wireless transmit/receive units (WTRUs) access to a core network of a cellular communication system. Within the radio access network, radio resources are divided into blocks of time (sub-frames) and frequency (frequency blocks).

(4) Transmission Time Interval (TTI) bundling in the uplink has been proposed to improve coverage for WTRUs near the cell edge. The solution is characterized by a single transport block

that is coded and transmitted in a set of consecutive subframes. A bundle is treated as a single resource, i.e., a single grant, and a single hybrid-Automatic Repeat Request (HARQ) acknowledgement is used for each bundle. The same HARQ process number is used in each of the bundled subframes. The HARQ Round Trip Time (RTT) is different than for the non-bundling case to reduce delays. The relation between subframe number and HARQ process number is unaffected for non-bundled subframes. Bundling can be applied to Frequency Division Duplex (FDD) as well as Time Division Duplex (TDD). For TDD, the bundling size needs to take the allocations of subframes to Uplink (UL) and Downlink (DL) into account.

(5) For an example in a LTE FDD system, a HARQ process and its different redundancy versions (RV) are bundled and transmitted in a fixed number of subframes, timeslots or applicable blocks of time or frequency and may be designated as N.sub.bundle and may also referred to as the TTI bundling value. For example, N.sub.bundle=4, is the current working assumption in 3GPP standards. A single transport block may be coded and transmitted in a set of consecutive subframes.

(6) Three alternatives for TTI bundles are shown in FIG. 1, FIG. 2 and FIG. 3, respectively. In Alternative 1, the timing relation between the last subframe in the TTI bundle and the transmission instant of the HARQ acknowledgement is identical to the case of no bundling. For the case of FDD, if the last subframe in a TTI bundle is subframe n, then the acknowledgement is transmitted in subframe n+4 and if the first subframe in a TTI bundle is subframe k, then any HARQ retransmissions begins in subframe $k+2*\text{HARQ_RTT}$. In Alternative 2, the timing relation between the first subframe in the TTI bundle and the transmission instant of the HARQ acknowledgement is identical to the case of no bundling. For the case of FDD, the HARQ acknowledgement is obtained from decoding the first subframe only. In Alternative 3, the timing relation between the last subframe in the TTI bundle and the transmission instant of the HARQ acknowledgement is identical to the case of no bundling. For the case of FDD, if the last subframe in a TTI bundle is subframe n then the acknowledgement is transmitted in subframe n+4

(7) Uplink TTI bundling may be activated and deactivated by radio resource controller (RRC) signaling. When switched on, TTI bundling applies to all uplink transmissions using PUSCH. To reduce the number of options and the associated testing, the number of configurations of the bundle size is minimized. Preferably only a single fixed value of the number of subframes in a bundle is specified. But there are several deficiencies with respect to this method such as but not limited to not providing criteria for triggering TTI bundling, no defined WRTU behaviors, no HARQ process related behaviors and no details related to Semi-Persistent Scheduling (SPS).

SUMMARY

(8) The present application discloses methods and apparatus for supporting TTI bundling in the uplink in evolved Universal Mobile Telecommunication System (UMTS) Terrestrial Radio Access (evolved UTRA) that includes for example new signaling mechanisms including the use of Physical Downlink Control Channel (PDCCH) and Medium Access Control (MAC) Control Element (CE) signaling, triggering criteria, activation and de-activation methods, wireless transmit/receive units (WTRU) behavior definitions, and TTI bundling with Semi-Persistent Scheduling.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more detailed understanding may be had from the following description, given by way of example in conjunction with the accompanying drawings wherein:

(2) FIG. 1 shows a Transmission Time Interval (TTI) bundling sequence;

(3) FIG. 2 shows an alternate TTI bundling sequence;

(4) FIG. 3 shows another alternate TTI bundling sequence;

- (5) FIG. 4 shows an example functional block diagram of a wireless transmit/receive unit (WTRU) and a base station;
- (6) FIG. 5(a) shows a standard Physical Downlink Control Channel (PDCCH) Downlink Control Information (DCI) format 0;
- (7) FIG. 5(b) shows a PDCCH DCI format 0 in accordance with an example embodiment;
- (8) FIG. 6(a) shows a PDCCH DCI format 0 in accordance with another example embodiment;
- (9) FIG. 6(b) shows a PDCCH DCI format 0 in accordance with yet another example embodiment;
- (10) FIG. 7 shows an example Information Element Medium Access Control (MAC) Configuration table in accordance with an example embodiment;
- (11) FIG. 8 shows an example TTI bundling diagram in accordance with an example embodiment;
- (12) FIG. 9 shows another exemplary TTI bundling diagram in accordance with an exemplary embodiment;
- (13) FIG. 10 shows an example hybrid-automatic repeat request (HARQ) process transmission after a TTI bundling deactivation in accordance with an example embodiment;
- (14) FIG. 11 shows an example TTI bundling diagram with an uplink grant in accordance with an example embodiment; and
- (15) FIG. 12 shows an example Information Element MAC Configuration table for Semi-Persistent Scheduling in accordance with an example embodiment.

DETAILED DESCRIPTION

- (16) When referred to hereafter, the terminology “wireless transmit/receive unit (WTRU)” includes but is not limited to a user equipment (UE), a mobile station, a fixed or mobile subscriber unit, a pager, a cellular telephone, a personal digital assistant (PDA), a computer, or any other type of user device capable of operating in a wireless environment. When referred to hereafter, the terminology “base station” includes but is not limited to a Node-B, a site controller, an access point (AP), or any other type of interfacing device capable of operating in a wireless environment.
- (17) Referring now to FIG. 4, there is shown an example functional block diagram **100** of a WTRU **110** and a base station **150** in communication with one another and may be configured to activate, deactivate, process and/or otherwise support Transmission Time Interval (TTI) bundling.
- (18) In addition to the components that may be found in a typical WTRU, the WTRU **110** includes a processor **115**, a transmitter **120**, a receiver **125**, and at least one antenna **130** to facilitate wireless communication. The processor **115** is configured to perform a method for activating, deactivating, processing and/or otherwise supporting TTI bundling in a wireless communication system in accordance with the present application. The receiver **125** and the transmitter **120** are in communication with the processor **115**. The antenna **130** is in communication with both the receiver **125** and the transmitter **120** to facilitate the transmission and reception of wireless data.
- (19) In addition to the components that may be found in a typical base station, the base station **150** includes a processor **155**, a receiver **160**, a transmitter **165**, and at least one antenna **170**. The processor **155** is configured to perform a method for activating, deactivating, processing and/or otherwise supporting TTI bundling in a wireless communication system in accordance with the present application. The receiver **160** and the transmitter **165** are in communication with the processor **155**. The antenna **170** is in communication with both the receiver **160** and the transmitter **165** to facilitate the transmission and reception of wireless data.
- (20) The following discussion presents signaling methods for TTI bundling, triggering criteria to activate/deactivate TTI bundling, activation and deactivation details, WTRU behavior upon activation and deactivation, processing issues, partial services with respect to TTI bundling and the use of Semi-Persistent Scheduling (SPS) with TTI bundling.
- (21) Signaling Methods and Information Provided
- (22) In addition to signaling using Radio Resource Control (RRC), exemplary methods for signaling and/or supporting TTI bundling include using the Physical Downlink Control Channel (PDCCH) and the Medium Access Control (MAC) Control Element (CE).

- (23) In an exemplary embodiment, PDCCH signaling is used as a method of signaling that supports TTI bundling. The PDCCH is used to carry downlink control information (DCI) such as scheduling grants, assignments, power-control commands, uplink scheduling grants, resource block allocations/assignments, and HARQ related information. The PDCCH carries multiple DCI formats depending on the type of required control information.
- (24) FIG. 5(a) and FIG. 5(b), show a DCI format 0 for a standard uplink grant and a DCI format 0 for an uplink grant that supports TTI bundling. The DCI format 0 includes multiple fields but for the purposes of the present description the relevant fields are Resource Block (RB) Allocation/Assignment; Modulation and Coding Scheme (MCS); and Demodulation Reference Symbol (DMRS).
- (25) Referring specifically to FIG. 5(b), the PDCCH can use one reserved or spare bit-field in either the RB allocation field, the MCS field or the DMRS field of the existing PDCCH DCI format 0 to indicate activation/de-activation of TTI bundling for the WRTU. In an illustrative embodiment, in PDCCH DCI format 0, a RB allocation field with “00000000000000” can be used to activate TTI bundling. The remaining fields of the disclosed PDCCH DCI format 0 can be used to indicate the TTI bundling configuration and necessary information for PUSCH transmission. The TTI bundling configuration information can include, but is not limited to activation/deactivation of TTI bundling for the WRTU; the number of bundled subframes (when and if applicable); and the maximum number of hybrid-automatic repeat request (HARQ) re-transmissions. Other TTI bundling configuration information can be provided as necessary such as, for example, the redundancy version (RV) value as further explained below.
- (26) A limited RB allocation and MCS may be used for the Physical Uplink Shared CHannel (PUSCH) when TTI bundling is applied, since the radio link condition may be very weak. The MCS and/or DMRS fields can be used to signal restricted RB allocation, MCS, the number of bundled TTIs or subframes and the TTI bundling configuration.
- (27) Referring now to FIG. 6(a) and FIG. 6(b), another example embodiment is shown using PDCCH signaling. In this example embodiment, the method uses PDCCH DCI formats 0 and 1A that are aligned (e.g., the same length) as shown in FIGS. 6(a) and 6(b), respectively. In this example embodiment, the PDCCH DCI format 0 adds an additional bit that provides the padding in order to have the same length as PDCCH DCI format 1A and the additional bit of DCI format 0 may be used to support TTI bundling switching and/or activation. When the TTI bundling bit is set to 1, it is an indication that TTI bundling is activated. The remainder of the PDCCH DCI format 0 can be used to indicate the rest of the TTI bundling configuration and information necessary for PUSCH transmission. As previously stated above, limited/restricted RB allocation and MCS may be used for PUSCH when TTI bundling is applied since the radio link condition may be very weak. The RB allocation, MCS and/or DMRS fields can be used to signal restricted RB allocation, MCS, the number of bundled TTIs or subframes and the TTI bundling configuration.
- (28) In another embodiment, a medium access control (MAC) control element (CE) is used to provide signaling for TTI bundling in the uplink. This can be a special MAC_CE used for TTI bundling signaling, e.g., activation/deactivation, along with providing the TTI bundling value or can be combined with other parameters in one MAC_CE. The TTI bundling value refers to the number of subframes, timeslots or other applicable units that are bundled together and treated as one unit. The MAC_CE header can indicate that this MAC_CE is TTI bundling signaling in the uplink. The contents of the MAC_CE can be the same as that proposed for using PDCCH signaling. Radio Resource Control (RRC) decides whether to use MAC_CE or PDCCH for TTI bundling signaling.
- (29) The MAC_CE can be used both for the configuration and reconfiguration of a TTI bundling value to the WTRU. The MAC_CE, to configure a TTI bundling value, can be sent at the time when a new service and/or connection is established or can be sent after the service and/or connection is established to reconfigure the TTI bundling parameters including a TTI bundling

value. The MAC_CE includes a TTI_Bundle_Value which can be set to 2, 4, 8 in this example, although other integer values could also be used in alternate embodiments.

(30) As stated above, the RV value may be provided in the signaling message. The RV value used for TTI bundling can be the same or different than the TTI bundling value. The RV values should correspond to the different number of TTIs that are to be used for the TTI bundle, e.g., RV values for 2, 4 or 8 bundled TTIs (as is currently proposed). It may be configured dynamically or semi-statically. When RV values used for TTI bundling are semi-static, they can be signaled during the configuration or re-configuration phase through RRC or MAC_CE signaling. If RV values for TTI bundling are dynamic, then they can be signaled during TTI bundling activation through MAC_CE or PDCCH signaling.

(31) When RRC message signaling is used to signal RV values, these values are contained in the IE MAC-Configuration (MAC-Configuration information element (IE)) which is used to specify the MAC layer configuration for data radio bearers as shown in FIG. 7. In an example embodiment, the RRC can signal all the RV values for all of the different TTI bundling values or in another embodiment, the RRC can signal RV values corresponding to the configured TTI bundling value. In the semi-static example embodiment, RVs for the configured TTI bundling value should be signaled.

(32) In an example embodiment for when MAC_CE signaling is used for RV values, the MAC_CE can signal all the RV values for all of the different TTI bundling values or in another embodiment, the MA_CE can signal RV values corresponding to the configured TTI bundling value. In the semi-static exemplary embodiment, RV values for the configured TTI bundling value should be signaled.

(33) Criteria to Trigger (Activate) TTI Bundling

(34) The criteria to trigger TTI bundling can be based on any combination of the following factors: Pathloss—this factor can indicate whether the WTRU is at the cell edge and if the WTRU is at cell edge this may mean the uplink channel quality is low; Signal to Noise and Interference ratio, e.g., SINR—this factor can indicate the signal quality and the interference level; number of retransmissions or number of NAKs, etc.—this factor can directly reflect the transmission quality; Quality of UL sounding channel—this factor can indicate whether the uplink channel has good quality; Quality of Service (QoS)—this indicator can reflect the priority and delay requirement of the service; power headroom; and uplink channel capacity. Analysis of the above factors can be used to determine whether to trigger TTI bundling in the following retransmissions.

(35) Activation Details

(36) Activation of TTI bundling can be included in the PDCCH when making the resource allocation for retransmission by the base station, e.g., an evolved NodeB (eNB). Activation can also or alternatively be accomplished using a MAC_CE. The activation can be signaled to a WTRU explicitly or implicitly.

(37) Explicit activation of TTI bundling can occur by using a one bit indication contained in the uplink grant of the PDCCH; a one bit indication at any other pre-defined position in the PDCCH; or by using a MAC_CE, where the MAC_CE can contain an indication to activate TTI bundling.

(38) Implicit activation of TTI bundling can occur if the uplink grant indicates that the resource allocation for retransmission of the initially transmitted packet is across several TTIs rather than one TTI. After the WTRU detects this information, the WTRU knows that this implies that the TTI bundling should be used in the following retransmissions. When semi-static TTI bundling is used, the WTRU implicitly knows from the RRC or MAC_CE because of the TTI bundling value that is sent for the retransmissions.

(39) There are two scenarios for uplink resource allocation for retransmissions using TTI bundling, thus different signaling methods should be used in PDCCH: Scenario 1: The same physical resources, (such as frequency bands and antenna, etc), transport format (such as MCS and power, etc), HARQ parameters (such as RV values) will be used for different TTIs in the same TTI bundle. In this scenario the uplink grant contained in PDCCH only needs to signal resource allocation

parameters for one TTI. The activation of TTI bundling will automatically imply to use the same resource allocation for the other TTIs (or subframes) in the TTI bundle. Scenario 2: Different physical resources, transport format and HARQ parameters will be used for different TTIs in the TTI bundle. Also the same RV values or different RV values can be applied to different TTIs in the TTI bundle. In this scenario the uplink grant contained in the PDCCH signals resource allocation parameters for all different TTIs in the TTI bundle. This can also implicitly indicate to the WTRU that TTI bundling is used for retransmissions.

De-Activation Details and WTRU Behavior

(40) De-activation of uplink TTI bundling and the corresponding WTRU behavior is now described. In an example embodiment, whenever the WTRU receives an ACK from a previous retransmission using TTI bundling, then the WTRU knows that the retransmission using TTI bundling has been successful and no more TTI bundling will be used for this packet. The WTRU should clear the HARQ buffer related to that HARQ process.

(41) When uplink TTI bundling is activated when at least one HARQ process is still active, (e.g., the WTRU is waiting for HARQ feedback or has a pending retransmission), the following provides example WTRU behaviors.

(42) In an example embodiment, the WTRU keeps the currently active HARQ process running until either a successful transmission occurs or the maximum number of retransmissions is reached. After that, TTI bundling is activated at the WTRU. From the time the RRC message is received to the time TTI bundling becomes effective at the WTRU's side, the base station, e.g., an eNB, should not schedule any uplink new data transmissions where the starting time falls in that range.

(43) In another example embodiment, the WTRU applies TTI bundling immediately. The TTI bundle size $N_{\text{sub.bundle}}$ determines the maximum number of HARQ processes, $N_{\text{sub.HARQ_bundle_max}}$, when TTI bundling is used. For example, for $N_{\text{sub.bundle}}=4$, $N_{\text{sub.HARQ_bundle_max}}$ equals to 4. If the number of currently active HARQ processes at the WTRU is more than $N_{\text{sub.HARQ_bundle_max}}$, it needs to be reduced to at most $N_{\text{sub.HARQ_bundle_max}}$. A pre-defined rule for which HARQ processes to keep or flush should be known to both the WTRU and the base station.

(44) In another example embodiment, the base station does not schedule any new data transmission for the WTRU for a fixed duration, say T . The duration T is long enough so that currently active HARQ processes at the WTRU will either succeed or fail after the maximum number of retransmissions when T expires. Then (after the duration of T), the base station will send signaling to the WTRU to activate uplink TTI bundling.

(45) When uplink TTI bundling is deactivated when at least one HARQ process is still active, (e.g., it waits for HARQ feedback or has a pending retransmission), the following provides example WTRU behaviors.

(46) In an example embodiment, the WTRU keeps the currently active HARQ process running until success or failure occurs after the maximum number of retransmissions is reached. After that, TTI bundling becomes disabled at the WTRU's side. From the time the RRC message is received to the time TTI bundling becomes disabled at the WTRU's side, the base station should not schedule any uplink new data transmission whose starting time falls in that range.

(47) In another example embodiment, the WTRU applies TTI bundling immediately. Within one bundle-mode HARQ round-trip-time (RTT), there are at most $N_{\text{sub.HARQ_bundle_max}}$ active HARQ processes at the WTRU. There are several example cases.

(48) In example case 1, if the time that TTI bundling deactivation signaling is received at the WTRU is early enough to allow the WTRU to stop transmission for the next bundled HARQ process, then the WTRU transmits the next (and subsequent) HARQ process in the non-bundled mode instead of bundle-mode, as shown in FIG. 8.

(49) In example case 2, if the time that TTI bundling deactivation signaling is received at the WTRU is not early enough to stop transmission for the next bundled HARQ process, say process n ,

then as shown in the exemplary example in FIG. 9, the WTRU should 1) transmit the HARQ process n in the bundled mode and 2) transmit the HARQ process $n+1$ (and subsequent ones) in the non-bundled mode instead of bundle-mode.

(50) In another example embodiment, the base station does not schedule any new data transmission for the WTRU for a fixed duration, say T . The duration T is long enough so that currently active HARQ processes at the WTRU will either succeed or fail after the maximum number of retransmissions when T expires. Then (after the duration of T), the base station sends signaling to the WTRU to deactivate uplink TTI bundling.

(51) When the WTRU switches from bundle mode to non-bundle mode (after receiving deactivation signaling), existing active HARQ processes in bundle-mode may need to be converted to non-bundle mode. Accordingly, HARQ processes converted from bundle-mode HARQ processes may be transmitted in the order of bundle-mode HARQ processes, as shown in FIG. 10.

(52) Synchronization Processing

(53) If uplink TTI bundling activation/deactivation is not properly configured or synchronized between the WTRU and base station, the problem needs to be detected and corrected. The following example methods for detection of the problem may be employed.

(54) In an example embodiment, the WTRU detects the (de)activation/synchronization problem. If the WTRU thinks it is configured for TTI bundling but receives an uplink grant conflicting with the current TTI bundling, it may regard the TTI bundling as not being configured correctly or the TTI bundling is not synchronized between the WTRU and base station. For example, the WTRU receives an uplink grant which schedules a new data transmission that will conflict with its TTI-bundled transmission, as shown in FIG. 11.

(55) In another example embodiment, the base station detects the activation/synchronization problem. For a WTRU that has been configured in the bundle-mode by the base station, if the base station detects that in the same bundle, one or several of the 2.sup.nd, 3.sup.rd, 4.sup.th HARQ transmissions has an energy level below a pre-determined threshold, then it will consider it as that WTRU is not (properly) configured in bundling mode.

(56) For a WTRU that has been configured in the non-bundle mode by the base station, if the WTRU is not scheduled to transmit in subsequent 2.sup.nd, 3.sup.rd, 4.sup.th sub-frames following an uplink data transmission in a set of RBs denoted by S , but the base station detects 1) energy levels at the set S (or a subset of S) in subsequent 2.sup.nd, 3.sup.rd, 4.sup.th sub-frames is above a (pre-determined) threshold even when set S (or a subset of S) is not allocated to other WTRUs; or 2) a collision with other WTRUs scheduled transmission on the set S (or a subset of S) in subsequent 2.sup.nd, 3.sup.rd, 4.sup.th sub-frames; then upon detecting the TTI bundling (de)activation/synchronization problem, the base station activates/deactivates the WTRU for uplink TTI bundling again.

(57) Partial Services

(58) In an example embodiment, when TTI bundling is configured for a WTRU, it is configured for all of its services. In another example embodiment, a partial (or mixed) TTI bundling mode can be configured. For illustrative purposes, a WTRU can be configured to use TTI bundling for low latency services, such as for example, voice or services with small radio link control (RLC) packet data unit (PDU) sizes. On the other hand, a WTRU can transmit in non-bundle mode for best effort services or services with large RLC PDU sizes.

(59) TTI Bundling and Semi-Persistent Scheduling.

(60) In an example embodiment, whether TTI bundling is used in Semi-Persistent Scheduling (SPS) mode may be predefined in the 3GPP standard or configured in a RRC or MAC_CE message signaling during the configuration phase, the reconfiguration phase, or both. When TTI bundling is used for SPS, the radio resource for retransmission may not be signaled by the base station via the PDCCH. However, this operation needs to be signaled by the base station so that the WTRU knows to perform retransmissions during SPS without waiting for reconfiguration signaling from the base

station The RRC or MAC_CE may signal to the WTRU whether the radio resource for retransmission will be reconfigured if TTI bundling is used for SPS.

(61) If no reconfiguration is indicated, then predefined frequency and resource blocks may be used for TTI bundling. The predefined radio resources may be the same for all bundled subframes, or the radio resources may vary from subframe to subframe according to the RRC or MAC_CE configuration. The variation may be a certain hopping pattern in the frequency domain. If it receives a NAK, the WTRU may not use the predefined radio resources for other uplink transmissions.

(62) When there is no need for reconfiguration for TTI bundling for SPS, the redundancy version (RV) value used for TTI bundling may be the same as is used for the initial transmission.

(63) When there is no signaling for retransmission resource allocation, and an uplink TTI-bundled retransmission collides with a subsequent initial TTI-bundled transmission, the initial transmission should take the higher priority. This means that, either the retransmission should be canceled or resource allocation is needed.

(64) When a RRC message is used to signal whether to use TTI bundling for SPS and whether to allow reconfiguration of resource for retransmissions, it can be contained in the Information Element (IE) MAC-Configuration, which is used to specify the MAC layer configuration for data radio bearers as shown in FIG. 12.

(65) When TTI bundling is used for SPS and the base station signals resource allocation for retransmissions, the PDCCH may only contain the resource allocation information for the first subframe. The WTRU may use the same resource allocation for other subframes in the same bundle.

(66) If the frequency and resource blocks are different subframe by subframe, then the resource allocation for retransmissions can be signaled for all the subframes used for TTI bundling.

(67) Activation and De-activation TTI Bundling for SPS via layer one Signaling, layer two Signaling, or both.

(68) Activation of UL TTI Bundling

(69) In an example embodiment, activation of uplink TTI bundling for SPS may be included in the PDCCH when making the resource allocation for retransmission by the base station. In another embodiment, activation can be done by using the MAC CE. The activation can be signaled to the WTRU explicitly or implicitly. Signaling for both methods is considered as follows.

(70) Explicit activation of TTI bundling can occur by using a one bit indication in the uplink scheduling grant carried on the PDCCH; a one bit, or one code-point, indication at any other pre-defined position in the uplink scheduling grant carried on the PDCCH; or by using the MAC_CE, where the MAC_CE can contain an indication to activate TTI bundling for SPS.

(71) Implicit activation of TTI bundling can occur if the uplink grant indicates that the resource allocation for retransmission of the initially transmitted packet is across several TTIs rather than one TTI. After the WTRU detects this information, the WTRU knows that this implies that TTI bundling should be used in the following retransmissions. When semi-static TTI bundling is used, the WTRU implicitly knows from the RRC or MAC_CE because of the TTI bundling value that is sent for the retransmissions.

(72) There are two scenarios for UL resource allocation for retransmissions using TTI bundling for SPS, thus different signaling methods should be used in PDCCH. Scenario 1: The same physical resources (such as frequency bands and antenna, etc.), transport format (such as MCS and power, etc.), HARQ parameters (such as RV values), may be used for different TTIs in the same TTI bundle. In this scenario, the uplink grant contained in the PDCCH may signal resource allocation parameters for the first subframe. The activation of TTI bundling will automatically imply the use of the same resource allocation for the rest of the TTIs in the TTI bundle. Scenario 2: Different physical resources, transport formats and HARQ parameters may be used for different TTIs in the TTI bundle. Either the same RV values or different RV values can be applied to different TTIs in

the TTI bundle. In this scenario, the uplink grant contained in PDCCH may signal resource allocation parameters for all different TTIs in the TTI bundle for SPS retransmission. This may implicitly indicate to the WTRU that TTI bundling is used for retransmissions.

De-Activation of Uplink TTI Bundling and WTRU Behavior

(73) De-activation of uplink TTI bundling and the corresponding WTRU behavior may be such that when the WTRU receives an ACK for a previous retransmission using TTI bundling, the WTRU knows that the transmission using TTI bundling is successful and no more TTI bundling will be used for the packet. The WTRU should clear the HARQ buffer related to that HARQ process.

(74) A method and apparatus for supporting uplink TTI bundling in LTE is provided. Described above are related method and apparatus for signaling, activation/deactivation and WTRU behavior. Signaling and wireless transmit receive unit behavior for transmission time interval bundling with semi-persistent scheduling in long term evolution is disclosed. Methods to support uplink TTI bundling in Semi-Persistent Scheduling (SPS) may include the use of RRC or MAC_CE. These signals may be used to inform the WTRU that the radio resources for retransmission will be reconfigured. Predefined frequency and resource blocks may be used for TTI bundling. The message can be contained in the Information Element (IE) MAC Configuration. Both activation and deactivation of TTI bundling are supported, and may be indicated explicitly or implicitly. Different signaling schemes may be used depending on whether the same physical resources are used for retransmissions.

(75) Although features and elements are described above in particular combinations, each feature or element can be used alone without the other features and elements or in various combinations with or without other features and elements. The methods or flow charts provided herein may be implemented in a computer program, software, or firmware incorporated in a computer-readable storage medium for execution by a general purpose computer or a processor. Examples of computer-readable storage mediums include a read only memory (ROM), a random access memory (RAM), a register, cache memory, semiconductor memory devices, magnetic media such as internal hard disks and removable disks, magneto-optical media, and optical media such as CD-ROM disks, and digital versatile disks (DVDs).

(76) Suitable processors include, by way of example, a general purpose processor, a special purpose processor, a conventional processor, a digital signal processor (DSP), a plurality of microprocessors, one or more microprocessors in association with a DSP core, a controller, a microcontroller, Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs) circuits, any other type of integrated circuit (IC), and/or a state machine.

(77) A processor in association with software may be used to implement a radio frequency transceiver for use in a wireless transmit receive unit (WTRU), user equipment (UE), terminal, base station, radio network controller (RNC), or any host computer. The WTRU may be used in conjunction with modules, implemented in hardware and/or software, such as a camera, a video camera module, a videophone, a speakerphone, a vibration device, a speaker, a microphone, a television transceiver, a hands free headset, a keyboard, a Bluetooth® module, a frequency modulated (FM) radio unit, a liquid crystal display (LCD) display unit, an organic light-emitting diode (OLED) display unit, a digital music player, a media player, a video game player module, an Internet browser, and/or any wireless local area network (WLAN) or Ultra Wide Band (UWB) module.

Claims

1. A wireless transmit/receive unit (WTRU), the WTRU comprising: a processor, wherein the processor is configured to: receive, via a physical downlink control channel (PDCCH), a downlink control information (DCI) associated with a repeated physical uplink shared channel (PUSCH) transmission, wherein the DCI indicates a repetition configuration, and indicates a number of time

- units associated with the repeated PUSCH transmission; determine hybrid automatic repeat request (HARQ) process information using the received DCI; and send the repeated PUSCH transmission in accordance with the number of time units, the repetition configuration, and the HARQ process information.
2. The WTRU of claim 1, wherein the processor is further configured to determine the number of time units indicated by the DCI.
 3. The WTRU of claim 1, wherein a time unit from the number of time units is a slot.
 4. The WTRU of claim 1, wherein a time unit from the number of time units is a subframe.
 5. The WTRU of claim 1, wherein the DCI further indicates a scheduled grant associated with the repeated PUSCH transmission.
 6. The WTRU of claim 1, wherein the DCI further indicates a modulation and coding scheme (MCS).
 7. The WTRU of claim 1, wherein the DCI further indicates a redundancy version (RV).
 8. The WTRU of claim 1, wherein the processor being is configured to send the repeated PUSCH transmission in accordance with the number of time units comprises the processor being configured to send the repeated PUSCH transmission using the number of time units.
 9. A wireless transmit/receive unit (WTRU), the WTRU comprising: a processor, wherein the processor is configured to: receive, via a physical downlink control channel (PDCCH), a downlink control information (DCI) associated with a repeated physical uplink shared channel (PUSCH) transmission, wherein the DCI indicates a repetition configuration, and indicates a number of time units associated with the repeated PUSCH transmission; determine hybrid automatic repeat request (HARQ) process information using the received DCI; determine the number of time units indicated by the DCI; and send the repeated PUSCH transmission in accordance with the number of time units, the repetition configuration, and the HARQ process information.
 10. The WTRU of claim 9, wherein a time unit from the number of time units is a slot.
 11. The WTRU of claim 9, wherein a time unit from the number of time units is a subframe.
 12. The WTRU of claim 9, wherein the processor being configured to send the repeated PUSCH transmission in accordance with the number of time units comprises the processor being configured to send the repeated PUSCH transmission using the number of time units.
 13. A method performed by a wireless transmit/receive unit (WTRU), the method comprising: receiving, via a physical downlink control channel (PDCCH), a downlink control information (DCI) associated with a repeated physical uplink shared channel (PUSCH) transmission, wherein the DCI indicates a repetition configuration, and indicates a number of time units associated with the repeated PUSCH transmission; determining hybrid automatic repeat request (HARQ) process information using the received DCI; and sending the repeated PUSCH transmission in accordance with the number of time units, the repetition configuration, and the HARQ process information.
 14. The method of claim 13, wherein the method further comprises determining the number of units indicated by the DCI.
 15. The method of claim 13, wherein a time unit from the number of time units is a slot.
 16. The method of claim 13, wherein a time unit from the number of time units is a subframe.
 17. The method of claim 13, wherein the DCI further indicates a scheduled grant associated with the repeated PUSCH transmission.
 18. The method of claim 13, wherein the DCI further indicates a modulation and coding scheme (MCS).
 19. The method of claim 13, wherein the DCI further indicates a redundancy version (RV).
 20. The method of claim 13, wherein the method further comprises sending the repeated PUSCH transmission in accordance with the number of time units by sending the repeated PUSCH transmission using the number of time units.
-